

# SENTIMENT ANALYSIS ON MENTAL HEALTH STATEMENTS

## *Project Report*

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## 1. Executive Summary

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This report presents the findings of a sentiment analysis project conducted on a dataset of 52,681 real-world mental health statements sourced from social media platforms including Reddit, Twitter, and Facebook. The objective was to build a machine learning model capable of automatically classifying text statements into one of seven mental health categories: Normal, Depression, Suicidal, Anxiety, Bipolar, Stress, and Personality Disorder.

Using Natural Language Processing (NLP) techniques and a Logistic Regression classifier, the model achieved an overall accuracy of 76% on unseen test data. The analysis revealed alarming patterns in the prevalence of mental health distress, particularly Depression and Suicidal ideation, which together accounted for nearly 50% of all statements in the dataset.

## 2. Project Overview

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### 2.1 Objectives

- Analyse real-world mental health text data to understand emotional and linguistic patterns
- Build a classifier that can automatically predict mental health status from text
- Identify key insights that could inform mental health awareness and intervention strategies
- Develop data visualisations that communicate findings clearly and compellingly

### 2.2 Dataset

The dataset used for this project was sourced from Kaggle and is titled "Sentiment Analysis for Mental Health" by Suchintika Sarkar. It is a carefully compiled combination of multiple mental health datasets drawing from Reddit, Twitter, and Facebook.

| Attribute            | Detail                               |
|----------------------|--------------------------------------|
| Dataset Name         | Sentiment Analysis for Mental Health |
| Source               | Kaggle (suchintikasarkar)            |
| Original Size        | 53,043 statements                    |
| After Cleaning       | 52,681 statements                    |
| Number of Columns    | 2 (statement, status)                |
| Number of Categories | 7                                    |

| Attribute | Detail                    |
|-----------|---------------------------|
| Data Type | Text (social media posts) |

## 2.3 Tools & Technologies

- Python 3 — core programming language
- Pandas & NumPy — data loading, cleaning, and manipulation
- NLTK (Natural Language Toolkit) — text preprocessing and stopwords removal
- Scikit-learn — TF-IDF vectorization and Logistic Regression modelling
- Matplotlib & Seaborn — data visualisation and charting
- WordCloud — word cloud generation per mental health category
- Jupyter Notebook — development environment

## 3. Methodology

### 3.1 Data Cleaning

The raw dataset contained 53,043 rows and 3 columns. The following cleaning steps were performed:

- Removed the unnecessary index column ("Unnamed: 0")
- Identified and removed 362 rows with missing statement values
- Reset the dataframe index after row removal

After cleaning, the dataset contained 52,681 rows across 2 columns with zero missing values.

### 3.2 Text Preprocessing

Raw text from social media is noisy and inconsistent. A custom cleaning function was applied to every statement in the dataset, performing the following steps in sequence:

- Converted all text to lowercase to ensure consistency
- Removed URLs and web addresses using regular expressions
- Removed all punctuation, numbers, and special characters
- Removed English stopwords (common filler words like "the", "and", "is") using NLTK's stopwords corpus

The cleaned text was stored in a new column called "cleaned\_statement" alongside the original for reference.

### 3.3 Feature Extraction — TF-IDF Vectorization

Machine learning models cannot process raw text directly — text must first be converted into numerical form. TF-IDF (Term Frequency — Inverse Document Frequency) was used for this purpose. TF-IDF assigns a numerical score to each word that reflects how important it is within a specific statement relative to the entire dataset. Words that appear frequently across all documents (and therefore carry little discriminatory value) receive lower scores, while words that are distinctive to particular categories receive higher scores. The top 5,000 most informative words were retained as features.

### 3.4 Model Training

The dataset was split into a training set (80%, 42,144 statements) and a testing set (20%, 10,537 statements) using a fixed random seed for reproducibility. A Logistic Regression classifier was trained on the TF-IDF transformed training data. Logistic Regression was selected for its interpretability, speed, and strong baseline performance on text classification tasks.

## 4. Results & Model Performance

### 4.1 Class Distribution

The dataset showed significant imbalance across the seven mental health categories:

| Mental Health Status | Number of Statements | Percentage |
|----------------------|----------------------|------------|
| Normal               | 16,343               | 31.0%      |
| Depression           | 15,404               | 29.2%      |
| Suicidal             | 10,652               | 20.2%      |
| Anxiety              | 3,841                | 7.3%       |
| Bipolar              | 2,777                | 5.3%       |
| Stress               | 2,587                | 4.9%       |
| Personality Disorder | 1,077                | 2.0%       |

### 4.2 Model Performance

The model was evaluated on 10,537 unseen test statements. Overall accuracy achieved was 76%. Detailed performance by category is shown below:

| Mental Health Status | Precision | Recall | F1-Score | Support |
|----------------------|-----------|--------|----------|---------|
| Normal               | 0.84      | 0.95   | 0.89     | 3,308   |

| Mental Health Status | Precision | Recall | F1-Score | Support |
|----------------------|-----------|--------|----------|---------|
| Anxiety              | 0.82      | 0.77   | 0.79     | 755     |
| Bipolar              | 0.85      | 0.74   | 0.79     | 527     |
| Depression           | 0.70      | 0.73   | 0.72     | 3,016   |
| Suicidal             | 0.68      | 0.64   | 0.66     | 2,158   |
| Stress               | 0.72      | 0.49   | 0.58     | 536     |
| Personality Disorder | 0.86      | 0.49   | 0.63     | 237     |

The model performed strongest on the Normal category (F1: 0.89), which had the most training data and the most distinctive language patterns. The weakest performance was on Stress (F1: 0.58), largely because stress-related language overlaps significantly with both Normal and Depression categories.

### 4.3 Confusion Matrix Analysis

The confusion matrix revealed that the most significant source of error was the overlap between Depression and Suicidal statements. Specifically, 555 Suicidal statements were misclassified as Depression, and 595 Depression statements were misclassified as Suicidal. This bidirectional confusion is linguistically expected, as suicidal ideation often develops from and closely mirrors depressive language patterns.

## 5. Key Insights

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### Insight 1: Depression is alarmingly prevalent

Depression (15,404 statements) and Suicidal ideation (10,652 statements) together account for nearly 50% of the entire dataset. When all distress categories are combined (Depression, Suicidal, Anxiety, Bipolar, Stress, and Personality Disorder), only 31% of statements were classified as Normal. Among people who openly express their mental state online, distress is far more common than wellbeing.

### Insight 2: Work and stress are deeply connected

In the Stress category, the word "work" was the single most dominant term after stopword removal. This points strongly to workplace pressure as a leading driver of stress and has significant implications for employee wellbeing programmes and organisational mental health policies.

### **Insight 3: Suicidal language closely mirrors depression**

The model's biggest confusion was between Depression and Suicidal statements. The linguistic overlap between these two categories suggests that suicidal ideation often develops from and sounds very similar to depressive language, reinforcing the critical importance of taking all expressions of depression seriously rather than dismissing them as "just sadness".

### **Insight 4: Bipolar disorder discussions are more clinical in nature**

Unlike other categories where people primarily described feelings and experiences, the Bipolar word cloud prominently featured words like "medication", "doctor", and "med". This suggests that people discussing Bipolar disorder are more likely to be engaged with the healthcare system and discussing treatment, rather than simply expressing raw emotion.

### **Insight 5: Loneliness is a recurring theme across multiple conditions**

Words such as "alone", "people", "friend", and "family" appeared consistently across Depression, Suicidal, and Personality Disorder categories. This suggests that relationship difficulties and feelings of social isolation are a common thread running through multiple mental health conditions, not unique to any one diagnosis.

### **Insight 6: First-person language dominates all categories**

The word "im" (I'm) was the most frequent word across nearly every mental health category. This confirms that these are deeply personal, first-person expressions of lived experience — making this dataset a uniquely honest and emotionally authentic window into how people actually feel when they believe they are speaking anonymously.

## **6. Recommendations**

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### **Recommendation 1: Treat Depression and Suicidal content as a connected risk spectrum**

Mental health platforms and content moderation systems should avoid treating Depression and Suicidal expressions as entirely separate categories. Given the strong linguistic overlap demonstrated in this analysis, automated detection tools should flag depressive language as a potential early indicator of suicidal risk and escalate accordingly.

### **Recommendation 2: Prioritise workplace mental health interventions**

The dominance of work-related language in stress statements suggests that organisations should invest more heavily in employee mental health support. Practical steps include flexible working arrangements, access to professional counselling, and reducing stigma around mental health disclosures in the workplace.

### **Recommendation 3: Address the data imbalance in future models**

Personality Disorder had only 1,077 statements in this dataset, leading to the weakest classification performance among minority categories. Future datasets should deliberately oversample underrepresented mental health conditions to enable more balanced and reliable classification across all categories.

### **Recommendation 4: Invest in social connection as a cross-cutting intervention**

Since loneliness and relational difficulties appear across Depression, Suicidal, and Personality Disorder categories simultaneously, community building and social support programmes could have broad mental health benefits that cut across multiple diagnostic categories at once.

### **Recommendation 5: Consider more advanced NLP models for future work**

While Logistic Regression with TF-IDF achieved a solid 76% accuracy as a baseline, more advanced approaches such as transformer-based models (e.g. BERT or RoBERTa) could potentially improve performance significantly, particularly on the harder-to-distinguish categories like Stress and Personality Disorder.

## **7. Conclusion**

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This project successfully demonstrated the application of Natural Language Processing and Machine Learning to the domain of mental health text classification. Starting from raw, unstructured social media text, the analysis pipeline covered data cleaning, text preprocessing, exploratory analysis, visualisation, model training, and evaluation — achieving a 76% classification accuracy across seven mental health categories.

Beyond the technical achievements, the most significant outcome of this project is the picture it paints of mental health in the digital age. The data reveals that among people who express themselves online about their mental state, distress is far more common than wellbeing — and that themes of loneliness, workplace pressure, and the blurring of depressive and suicidal language are patterns that deserve urgent attention from mental health professionals, platform designers, and policymakers alike.

This analysis represents a strong foundation for future work in computational mental health, and demonstrates the genuine potential of data analytics to contribute meaningfully to one of the most important public health challenges of our time.

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